

Intelligent Detection of Student Classroom Behavior Based on an Improved YOLOv8 Model

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Abstract: This paper proposes an intelligent detection method of student classroom behavior based on an improved YOLOv8 model. First, a dataset comprising five categories of classroom misbehaviors, i.e., “using mobile phones”, “sleeping on the desk”, “looking down”, “whispering” and “looking around” is constructed. Second, an improved YOLOv8 model is designed to detect the student classroom behavior. To improve the model’s robustness in complex scenes, a spatial adaptive feature modulation network is introduced, leveraging multi-scale feature fusion. Then, the proposed method is validated on the constructed dataset, and the results demonstrate that the proposed method has improvement in recognition accuracy, i.e., increasing the mean average precision (mAP@0.5) to 81.7%. Furthermore, a visual system interface is developed using PyQt5. Leveraging integrated real-time video analysis and behavioral metrics, this work facilitates optimized teaching management and contributes to the development of a smart classroom ecosystem.

Key Words: Classroom behavior detection, improved YOLOv8 model, educational intelligence, visual system interface, multi-scale feature fusion

1 Introduction

With the rapid advancement of artificial intelligence technology, various industries are exploring the integration of advanced and efficient artificial intelligence solutions. The “Education Informatization 2.0 Action Plan” calls for building a new “Internet + Education” ecosystem. It also highlights using new technologies like artificial intelligence to transform education models [1]. Furthermore, “China’s Education Modernization 2035” lists speeding up educational reform in the information age as a key task. This provides strong policy support for the education sector’s digital transformation [2].

The classroom environment is crucial for ensuring teaching quality and student learning efficiency. Well-regulated classroom discipline not only enhances knowledge transfer but also serves as a vital platform for cultivating students’ concentration and self-management abilities. However, real-world teaching scenarios frequently encounter classroom misbehaviors such as students using mobile phones, sleeping, and engaging in unauthorized conversations. These actions not only distract individual focus but also create a chain effect, persistently disrupting the learning environment. Traditional classroom supervision, relying primarily on manual observation by teachers, suffers from limitations such as significant blind spots and difficulties in quantification, hindering precise and real-time behavior management. Consequently, classroom behavior detection technology has become a prominent research area in the effort to integrate classrooms with digital intelligence.

The earliest research on automatic face recognition originated in the 1960s [3]. By the late 1980s, the field of automatic face recognition mainly had two research directions: template-based recognition methods and feature-

based recognition methods. Among them, template-based recognition methods included pyramid machines and neural network methods. These template-based methods identify faces by matching them with stored templates based on similarity. Feature-based recognition methods included deformable templates. These feature-based methods identify faces by extracting and matching key facial features, which makes the approach more sensitive and flexible to local variations. In the statistical learning recognition framework, the Eigenface method is proposed, in which principal component analysis is employed for face recognition [4]. While the Viola-Jones algorithm significantly enhances face detection speed by integrating AdaBoost, integral images, and cascaded classifiers [5].

In recent years, deep learning technique has developed rapidly. In the 2012 ImageNet image classification competition, the AlexNet deep convolutional neural network was designed by Alex Krizhevsky and Geoffrey Hinton et al., which was achieved a decisive victory [6]. Its key advantages mainly lie in the deep convolutional neural network architecture, GPU parallel computing, ReLU activation function and mini-batch stochastic gradient descent, and it also uses Dropout layers to mitigate overfitting. Human action recognition mainly involves two processes: feature representation, action recognition and understanding. Among them, feature representation extracts features that can represent the key information of the video from the video data, this process plays a crucial role in the entire recognition process, and the quality of the features directly affects the final recognition result, in addition, the action recognition and understanding stage takes the feature vectors obtained in the previous stage as input and uses machine learning algorithms to learn, and inputs the feature vectors obtained in the testing process or application scenarios into the model obtained in the above process for type recognition [7]. Later, a new detection method named YOLO was proposed. From then on, the series of YOLO models have led to significant im-

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provements in both speed and accuracy in object detection [8].

In general, deep learning-based object detection can be divided into two categories: two-stage detection and one-stage detection. Two-stage detection is a “coarse-to-fine” process, while the other is an end-to-end “one-step completion”. Usually, one-stage detection attempts to directly classify each region of interest as background or target object. It can directly give the category probability and position coordinate value of the object through only one stage. Typical examples include the YOLO series. In 2018, Redmon and Farhadi used YOLOv3 with optimized feature extraction and multi-scale detection to improve the recognition of small objects, especially rear-row faces in classroom activity detection. Next in 2020, Ultralytics proposed YOLOv5 to improve both detection speed and accuracy, leading to its widespread adoption in student classroom behavior recognition. Then in 2023, Ultralytics designed YOLOv8, which was built on YOLOv5 with further optimizations for speed-accuracy balance, to create an efficient and robust framework for real-time student classroom behavior detection [9]. YOLOv8 is boasting higher detection accuracy and faster inference speed. It supports multi-task learning, such as object detection and pose estimation, and performs excellently in scenarios like real-time video analysis, autonomous driving, medical imaging and agricultural monitoring.

Based on the above analysis of the YOLO series, this paper selects YOLOv8 model to implement intelligent monitoring of student classroom behavior, hoping to provide assistance for the construction of future smart classrooms driven by big data. Currently, big data-driven smart classrooms are an inevitable trend in the future of education [10]. Observing student classroom behavior mainly relies on teachers’ manual supervision and video analysis. However, this approach distracts teachers’ attention, reduces teaching efficiency, and makes it difficult to comprehensively and accurately record all students’ behaviors. Therefore, there is a growing expectation for the application of deep learning and computer vision techniques to analyze student behavior in the classroom. Consequently, the automatic detection and analysis of students’ classroom behavior using computer-aided instruction has become a prominent research topic in smart education [11, 12].

This paper proposes a computer vision-based method for intelligent student behavior detection, aiming to overcome the inefficiency and subjectivity of manual classroom management. The main contributions of this paper consist of the following aspects:

1. A dataset containing five types of classroom misbehaviors is constructed, which comprises 1089 images from various classroom environments, annotated using the labeling tool, and divided into training and validation sets.
2. An improved YOLOv8 model is constructed, in which a spatial adaptive feature modulation network is introduced to enhance multi-scale feature fusion capabilities. This improves the model’s detection performance in complex scenarios and enables real-time recognition and statistical analysis of classroom behavior.

The following structure of this paper is organized as fol-

lows. In Section 2, the dataset is established. In Section 3, the main improved model is designed. In Section 4, simulation verification is conducted. Finally, some conclusions of this work are given in Section 5.

2 Dataset Construction

High-quality data allows models to accurately fit the association between features and targets [13]. Therefore, a large amount of image and video data depicting various student behaviors in the classroom should be collected to train models with high recognition accuracy and strong scene generalization capabilities. In this section, a dataset for student learning behavior detection in natural scenes is constructed to transform the abstract problem of “classroom management” into a quantifiable object detection task, a clear and evaluable optimization direction is provided for subsequent model design.

Considering the diverse and complex nature of student behavior in the classroom, five common behavioral characteristics have been selected for analysis, i.e., using mobile phones, sleeping on the desk, bowing the head, whispering, and looking around. Detailed description is shown in Table 1.

Table 1: Student Misbehavior Recognition

Student Misbehavior	Description
Using a mobile phone	Using mobile phones in various postures
Sleeping on the desk	Lying on their desks
Bowing the head	Bow their heads with a large angle
Whispering	Tilt their heads and are in a speaking state
Looking around	Turn their heads or bodies significantly

The data sources are mainly twofold: obtained from online platforms (such as the Mota community and CSDN) and a self-made dataset, consisting of simulated classroom misbehavior videos filmed by our school’s students.

In this paper, we construct a dataset of 1089 images, containing 4374 annotations of “using mobile phone”, 1252 of “sleeping on desk”, 4071 of “head down”, 970 of “whispering”, and 1711 of “looking around”. Captured from top-down and horizontal perspectives, these images encompass diverse classroom environments and lighting conditions, thereby ensuring the model’s generalization ability. In the following, the above annotations will be abbreviated as “phone”, “lean”, “bow”, “talk”, “turn head”, respectively.

Next, the images are cropped and filtered to highlight feature quality, which enhances the model’s robustness during training, enabling it to better adapt to recognition in diverse environments and conditions. Finally, based on the evaluation criteria for students’ classroom misbehaviors, the rectangular annotation of the image data using the labeling toolkit is proposed.

Upon completion of the annotation process, the system generates a JSON-formatted label file, which is then converted into a TXT format file for YOLO training. The annotated dataset is subsequently split into training and validation sets with a 7:3 ratio. These sets are then organized into the following directories: training images, validation images, training labels, and validation labels, facilitating their use during subsequent YOLO model training.

3 Main Results

3.1 YOLOv8 Model Structure

Compared with previous YOLO models, YOLOv8 offers significant improvements in both speed and accuracy, especially excelling in small object detection and complex scenes. Its enhanced network architecture and optimized training strategies make it more efficient for real-time detection tasks, while also increasing robustness. YOLOv8 has strong performance in its efficiency and stability, hence, this paper selects YOLOv8 as the benchmark network [14].

The YOLOv8 structure is mainly composed of two parts: a Backbone for feature extraction and a Head for prediction output. These components synergize through multi-level feature fusion, ensuring real-time performance without compromising detection accuracy. The entire data flow progresses layer by layer from input to output, showcasing a concise and efficient structural design that reflects careful consideration of the balance between computational resources and performance. The Backbone employs a top-down, level-by-level transmission structure, primarily consisting of alternating convolutional (Conv) and C2f layers. The Head begins with an Upsample layer, which receives data from the SPPF layer located at the Backbone's end.

3.2 Model Improvement

Object detection task involves recognizing both the position of objects and categorizing each object within a given image. The YOLO series employs a single-stage approach to address object detection tasks, enabling one-shot detection. This single-stage algorithm also significantly improves computational speed [15–17]. The performance of YOLO models can be enhanced through techniques such as multi-scale training and testing, feature pyramid networks, multi-task learning, and multi-modal information fusion [18]. A direct method to improve YOLO is PDNet, which predicts an object's class and boundary location separately from different locations to improve prediction accuracy [19].

While YOLOv8s exhibits strong performance on certain datasets, its generalization capabilities are limited. In particular, it struggles with recognizing students' behavior under occlusion, and its accuracy can still be improved. To address these issues, in this work a spatial adaptive feature modulation (SAFM) is considered to enhance the model. The SAFM is proposed by Sun Long et al., it is a deep network designed for image super-resolution tasks [20]. Its structure is shown in Fig.1. By combining local and global features, SAFM processes and fuses data through its components, stabilizes training, and captures details to achieve high-quality image reconstruction.

The SAFM module uses a multi-scale feature interaction mechanism to adaptively integrate visual information, it captures feature information at different levels by employing convolutional kernels or pooling layers of varying sizes. And first multi-angle features are extracted as

$$F_{ms} = \{F_{s1}, F_{s2}, F_{s3}\} \quad (1)$$

where F_{s1} , F_{s2} , and F_{s3} represent feature maps at different scales.

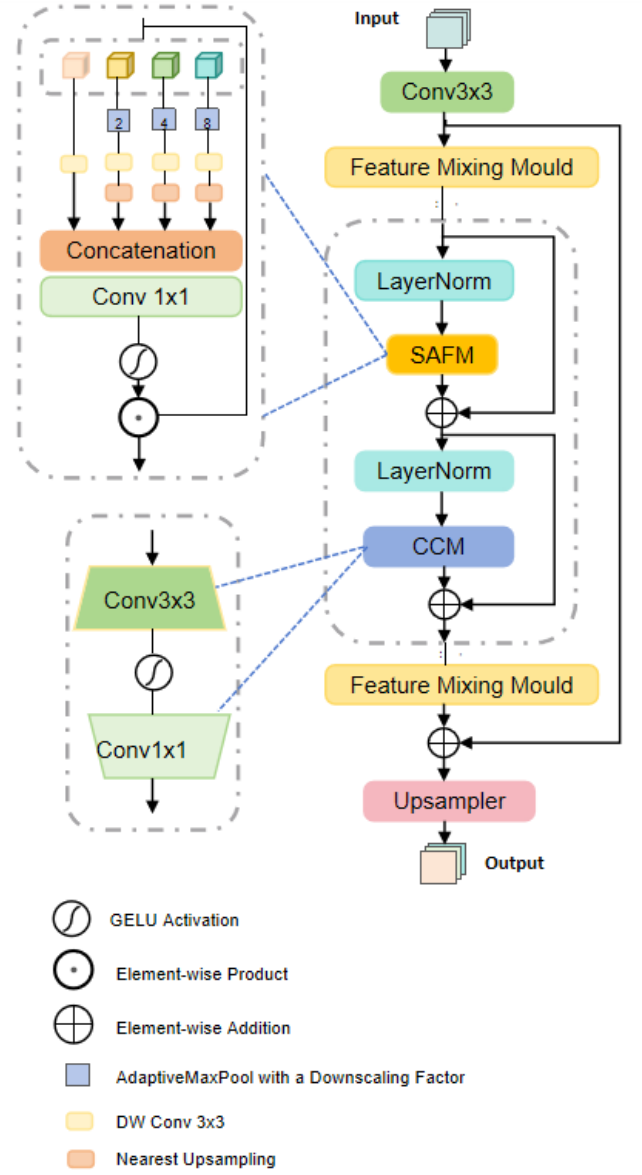


Fig. 1: SAFM Architecture Diagram

Feature fusion is subsequently performed as

$$F_{fused} = \text{Concat}(F_{s1}, F_{s2}, F_{s3}) \quad (2)$$

By concatenating feature maps of different scales along the channel dimension and integrating multi-scale feature information, the formula allows the network to consider both details and global context, thereby enriching the feature representation.

Finally, the features are modulated as

$$F_{modulated} = F_{fused} \odot \sigma(W \cdot F_{fused}) \quad (3)$$

where σ , W and $\odot$ denote the activation function, weight matrix and element-wise multiplication.

First, the fused feature map F_{fused} undergoes a linear transformation via the weight matrix W . Then, non-linearity is introduced through the activation function σ . Finally, element-wise multiplication ($\odot$) is used to multiply the original fused feature map by the transformed feature map, achieving adaptive modulation of features. This modulation

mechanism allows the network to dynamically adjust the importance of different parts of the input features.

The convolutional channel mixer (CCM) module employs a three-stage structure, serially connected by a convolutional layer of size 3, a GELU activation layer, and a convolutional layer of size 1. By sequentially performing spatial feature extraction, non-linear transformation, and channel dimension compression, the module achieves lightweight feature optimization while preserving the input resolution.

The module operates by first applying convolution to the input feature maps, which is formulated as

$$F_{\text{conv}} = \text{Conv}(F_{\text{input}}) \quad (4)$$

where the convolution operation performed on the input feature map F_{input} . Convolutional layers are the fundamental building blocks of Convolutional Neural Networks, which extract local features and generate a new feature map F_{conv} .

Then the GELU activation function is applied to introduce non-linearity as

$$F_{\text{activated}} = \text{GELU}(F_{\text{conv}}) \quad (5)$$

where $F_{\text{activated}}$ represents the application of the GELU activation function to the convolved feature map F_{conv} . GELU is a commonly used activation function with the representation as

$$\text{GELU}(x) = x \cdot \Phi(x) \quad (6)$$

Finally, the channel dimension is reduced through a convolutional layer of size 1, as shown as

$$F_{\text{compressed}} = \text{Conv}(F_{\text{activated}}) \quad (7)$$

where $F_{\text{compressed}}$ describes channel compression of the activated feature map $F_{\text{activated}}$, achieved through a convolutional layer of size 1.

A convolutional layer of size 1 is a specialized convolution operation that preserves the spatial dimensions of the feature map while adjusting the number of channels, enabling dimensionality reduction or expansion.

The SAFM mechanism can be applied to student classroom behavior detection. It effectively fuses multi-scale features and employs dynamic spatial modulation to focus on key regions. Furthermore, the CCM within this mechanism captures local details such as gestures and expressions. And the lightweight design of the SAFM structure, with its reduced parameters and computational cost, makes it well-suited for real-time video stream processing and facilitates easy deployment on classroom monitoring devices. The model's stability across diverse classroom environments is also enhanced by its adaptive multi-scale feature extraction and normalization strategy. This achieves a favorable balance between efficiency and accuracy, offering a practical and reliable improvement for real-time classroom behavior analysis.

4 Simulation Verification

4.1 Experimental Setup

As shown in Fig.2, a dataset is established in the experiment. For the YOLOv8s model, the depth multiple is set to 0.33, and the channel multiple to 0.50. The pre-labeled materials are then used to train the model via a custom-written



Fig. 2: The Behavior dataset is shown above

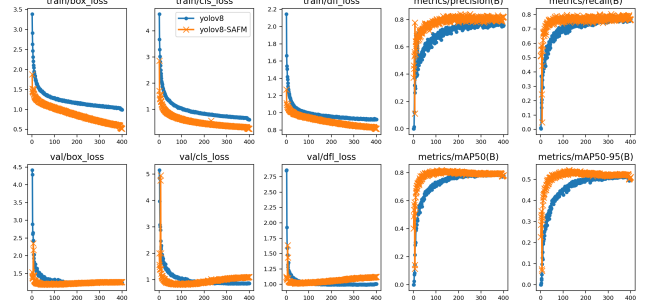


Fig. 3: Performance comparison between YOLOv8 and YOLOv8-SAFM models

train file, where training parameters are configured. Specifically, the epoch is set to 400, although the model typically converges within 300 rounds. With CUDA pre-installed, the device is set to 0 to leverage the GPU for training.

4.2 Model Results and Analysis

To demonstrate the effectiveness of the proposed method, it is compared with YOLOv8 model. Through experimental comparison, YOLOv8-SAFM shows a faster decline in loss functions during both training and validation processes. Moreover, it outperforms the original YOLOv8 model in terms of final precision, recall, and mean Average Precision (mAP), demonstrating superior performance, as illustrated in Fig.3.

By comparing the performance of the baseline model and the improved model on the behavior dataset, the results show that the improved model has shown improvements in multiple key metrics. Specifically, the mAP@0.5 of the improved model increased from 0.809 to 0.817, representing an absolute increase of 0.8 percentage points, the mAP@[0.5:0.95] also increased from 0.526 to 0.544, achieving an absolute improvement of 1.8 percentage points. Detailed comparison results are presented in Table 2. This performance optimization is primarily attributed to the improvement in model design, which significantly enhanced the model's accuracy in recognizing the four behaviors: "phone", "lean", "bow", "talk" and "turn head" thereby meeting the basic standards for high-precision detection. And the results of system identification are shown in Fig.4.

4.3 Classroom Behavior Detection Application Design

PyQt5 is a GUI development toolkit built on the Python language. By seamlessly integrating the Qt framework with

Table 2: The Performance of the Baseline model and the improved model

Methods	P	R	mAP 0.5	mAP 0.5:0.95
YOLOv8	0.785	0.768	0.809	0.526
YOLOv8s-SAFM	0.809	0.769	0.817	0.544



Fig. 4: The recognition results of the overall algorithm for the Behavior dataset

Python, it combines the robustness of Qt with the simplicity of Python, creating an ideal cross-platform development environment.

The system interface is simple and intuitive, adopting the design of left and right split screen. The display area on the left can dynamically present the system analysis picture, and the output window of system text information is below it. Five function keys on the right cover the main functions of the system. The interface of the system running is shown in Fig.5.

Users can click the “Select File” button to import classroom video and images. The system supports processing videos in MP4, AVI and MOV formats as well as pictures in JPG and PNG formats. After clicking “Start Recognition”, the system first processes the input video and images to enhance model recognition accuracy. It then activates the pre-trained model to identify students’ head posture, limb movements, and other features in the data. This enables precise detection of classroom misconduct behaviors such as using mobile phones or sleeping at desks, with detection results displayed in the text output window below.

After completing the recognition process, click the “Save Results” function key to save the recognition result im-



Fig. 5: Class behavior analysis system operation interface

age to a designated location for later review and traceability. Clicking the “Classroom Data Extraction” function key generates corresponding statistical data: for images, it produces a category-quantity bar chart, for videos, it generates a category-quantity-time line chart. The classroom data images generated from video and image recognition are shown in Fig.6.

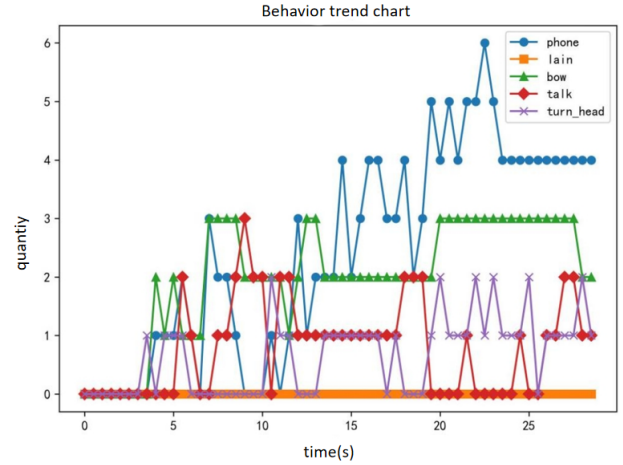


Fig. 6: Category-quantity-time line chart

The system achieves full-process coverage from data collection to in-depth analysis. The generated quantitative indicators assist teachers in optimizing instructional designs and support educational researchers in studying teaching behavior patterns, providing reliable technical backing for building a smart classroom ecosystem.

5 Conclusion

This paper proposes a student classroom behavior detection system based on computer vision and artificial intelligence technologies, which can effectively detect inattentive behaviors of students in class, such as using mobile phones, sleeping on the desk, looking down, whispering and looking around. The proposed behavior detection algorithm is built based on an improved YOLOv8 model. To enhance the perception ability of the model, a SAFM is introduced, which incorporates adaptive max-pooling to realize flexible regulation of the receptive field range, enabling better perception of the spatial features of target objects. Finally the PyQt5 toolkit is also used to implement a monitoring platform that is simple to operate, easy to use, and has good visual effects.

The currently used dataset has certain limitations in terms of scene coverage, behavior categories, and sample quantity, making it unable to be applied to all classroom environments. In the future, it is necessary to expand the data sources, including collecting classroom behavior data from more classroom environments, lighting conditions, and crowds, to form a more representative classroom behavior dataset and provide possibilities for the generalization of the system.

References

- [1] Yan S, Yang Y. Education informatization 2.0 in China: motivation, framework, and vision. *ECNU Review of Education*, 2021, 4(2): 410-428.

- [2] Zhu Y. New national initiatives of modernizing education in China. *ECNU Review of Education*, 2019, 2(3): 353-362.
- [3] Gupta O P, Agrawal A P, Pal O. A study on evolution of facial recognition technology// 2023 International Conference on Disruptive Technologies (ICDT). IEEE, 2023: 769-775.
- [4] Turk M. Over twenty years of eigenfaces. *ACM Transactions on Multimedia Computing, Communications, and Applications (TOMM)*, 2013, 9(1s): 1-5.
- [5] Viola P, Jones M. Rapid object detection using a boosted cascade of simple features// Proceedings of the 2001 IEEE Computer Society Conference on Computer Vision and Pattern Recognition. CVPR 2001. IEEE, 2001, 1: 1-1.
- [6] Krizhevsky A, Sutskever I, Hinton G E. ImageNet classification with deep convolutional neural networks. *Communications of the ACM*, 2017, 60(6): 84-90.
- [7] Kong Y, Fu Y. Human action recognition and prediction: A survey. *International Journal of Computer Vision*, 2022, 130(5): 1366-1401.
- [8] Redmon J, Divvala S, Girshick R, et al. You only look once: unified, real-time object detection// Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. 2016: 779-788.
- [9] Cao Y, Cao Q, Qian C, et al. YOLO-AMM: A real-time classroom behavior detection algorithm based on multi-dimensional feature optimization. *Sensors*, 2025, 25(4): 1142.
- [10] Singh H, Miah S J. Smart education literature: a theoretical analysis. *Education and Information Technologies*, 2020, 25(4): 3299-3328.
- [11] Zhou J, Ran F, Li G, et al. Classroom learning status assessment based on deep learning. *Mathematical Problems in Engineering*, 2022(1): 7049458.
- [12] Hu M, Wei Y, Li M, et al. Bimodal learning engagement recognition from videos in the classroom. *Sensors*, 2022, 22(16): 5932.
- [13] Liu Q, Jiang R, Xu Q, et al. Yolov8n.bt: research on classroom learning behavior recognition algorithm based on improved YOLOv8n. *IEEE Access*, 2024.
- [14] Han L, Ma X, Dai M, et al. A WAD-YOLOv8-based method for classroom student behavior detection. *Scientific Reports*, 2025, 15(1): 9655.
- [15] Wang S, Xu M, Sun Y, et al. Improved single shot detection using DenseNet for tiny target detection. *Concurrency and Computation: Practice and Experience*, 2023, 35(2): e7491.
- [16] Diwan T, Anirudh G, Tembhurne J V. Object detection using YOLO: challenges, architectural successors, datasets and applications. *Multimedia Tools and Applications*, 2023, 82(6): 9243-9275.
- [17] Wang P, Sun X, Diao W, et al. FMSSD: feature-merged single-shot detection for multiscale objects in large-scale remote sensing imagery. *IEEE Transactions on Geoscience and Remote Sensing*, 2019, 58(5): 3377-3390.
- [18] Zhao Z Q, Zheng P, Xu S, et al. Object detection with deep learning: a review. *IEEE Transactions on Neural Networks and Learning Systems*, 2019, 30(11): 3212-3232.
- [19] Yang L, Xu Y, Wang S, et al. PD Net: toward better one-stage object detection with prediction decoupling. *IEEE Transactions on Image Processing*, 2022, 31: 5121-5133.
- [20] Sun L, Dong J, Tang J, et al. Spatially-adaptive feature modulation for efficient image super-resolution// Proceedings of the IEEE/ CVF International Conference on Computer Vision. 2023: 13190-13199.